

The Data Drivers Present



EV Charging Points in Singapore

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Agenda

Project Overview

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Data Source and Attributes

Methodology & Tools

Data Analysis & Visualisation

Challenges & Limitations

Areas for Improvement

Conclusion

Project Overview

- Accessible charging infrastructure is essential for EV adoption.
- However, charging-point data can be difficult for prospective EV buyers to interpret.
- This project transforms complex charging-point data into structured information, providing clearer insights into Singapore's EV charging infrastructure.

Objectives

Extract, clean and restructure EV charging-point data from LTA DataMall

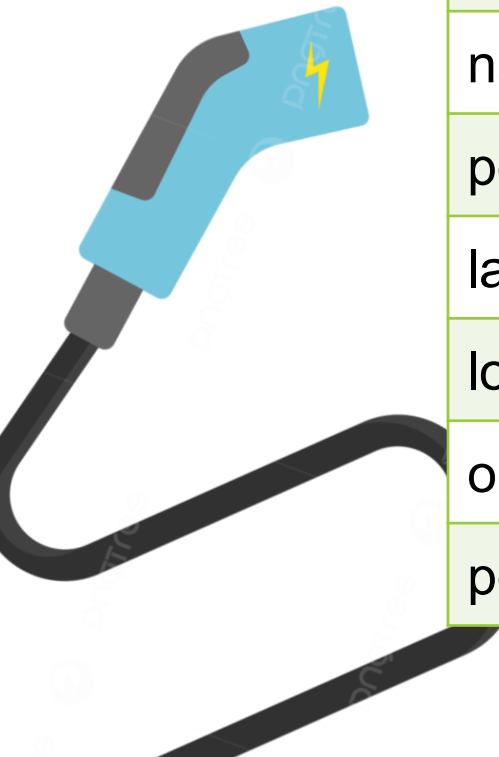
Design and load the cleaned data into a structured PostgreSQL database

Analyse charging-point operators, power ratings and regions

Identify data challenges, limitations and opportunities for future improvement

Data Source 1: LTA DataMall – Location and Operator Attributes

<https://datamall.lta.gov.sg/content/datamall/en/dynamic-data.html>



Attributes	Description
address	Address of charging station
name	Name of charging station
postalCode	Postal code of charging station
latitude	Latitude map coordinates of charging station
longitude	Longitude map coordinates of charging station
operator	Charging operator of the charger
position	Position of the charger

Data Source 1: LTA DataMall – Charger and Plug Attributes

Attributes	Description
plugType	Plug type of the charging point
current	AC/DC
powerRating	Charging power of the charging point (kW)
price	Charging price
priceType	Pricing basis (\$/h, \$/kWh)
evCpId	Connector ID. Assigned by LTA during charger registration
status	Status of the charging point



Data Source 2: GitHub – Postal Code and Region Attributes

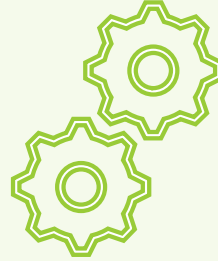
https://github.com/mhiyer/map_singapore_postal_codes_to_regions

Attributes	Description
District	A unique ID given to each cluster of towns
Area	A cluster of towns sharing the same or sequential postal code
Postal Code	Postal sector (contains only the first 2 digit of postal code)
Region	North, North-East, East, West, Central

Methodology



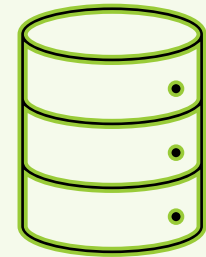
1. Extract Data



2. Flatten & Transform



3. Clean & Validate



4. Design & Load



5. Analyse Data



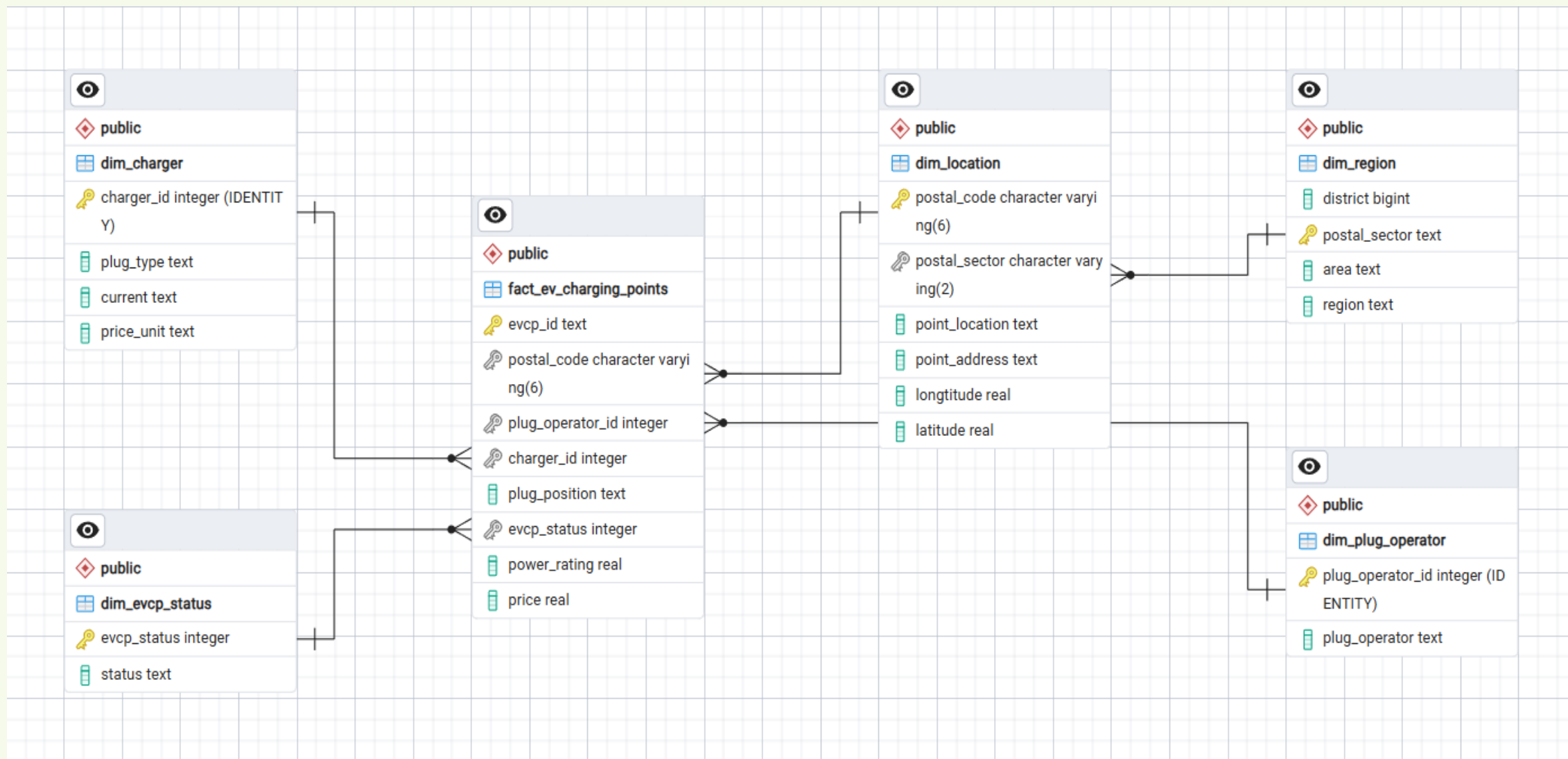
6. Visualise Findings

Tools

- Python and its libraries: NumPy, pandas, requests, SQLAlchemy
- JSON data format
- Jupyter Notebook
- PostgreSQL
- Git and GitHub
- Power BI



Database Design



Number of EV Charging Points by Operator

EV Charging Point Operator

SP MOBILITY PTE. LTD.
CHARGE+ PTE. LTD.
COMFORTDELGRO ENGIE PTE. LTD.
SHELL SINGAPORE PTE. LTD.
STRIDES YTL PTE. LTD.
VOLT SINGAPORE PTE. LTD.
MNL SOLUTIONS PTE. LTD.
GREAT CHARGE PTE. LTD.
FASTPARKNCHARGE PTE. LTD.
EV MOBILITY PTE. LTD.
TESLA MOTORS SINGAPORE PRIVATE LIM...
KED ENERGY PTE. LTD.
NOVOWATT PTE. LTD.
EIGEN ENERGY PTE. LTD.
BUSWAYS PTE. LTD.
ST ENGINEERING URBAN SOLUTIONS LTD.
EVONE CHARGING PTE. LTD.
CITY ENERGY GO PTE. LTD.
SOLATEKS PTE. LTD.
UP SOLUTIONS PTE. LTD.
AIRETEC PTE. LTD.
ENEREADY PTE. LTD.
UNION GAS PTE. LTD.
WES CARES ENERGY PTE. LTD.
G. TECH PTE. LTD.
ELECTRIC PTE. LTD.
IWOW TECHNOLOGY LIMITED
P-Parking International Pte Ltd
ALPINA ENERGY PTE. LTD.

0

500

1,000

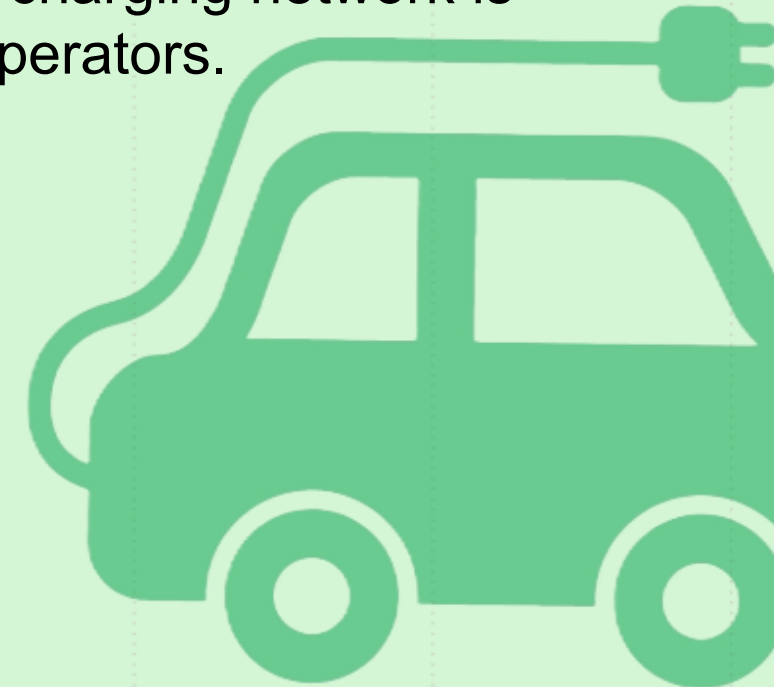
1,500

2,000

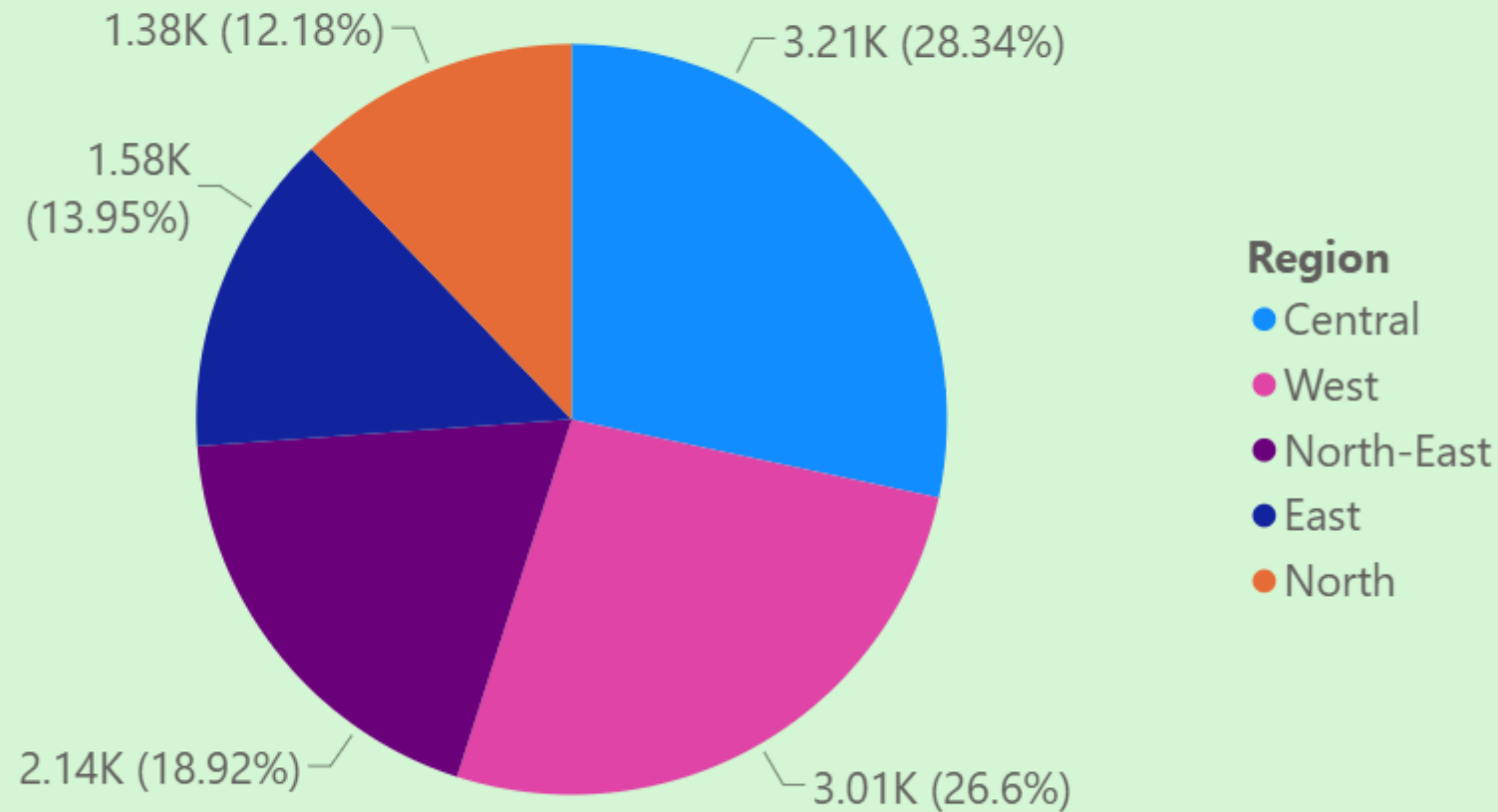
2,500

3,000

SP Mobility and Charge+ operate the most charging points. Overall, Singapore's charging network is dominated by a few major operators.

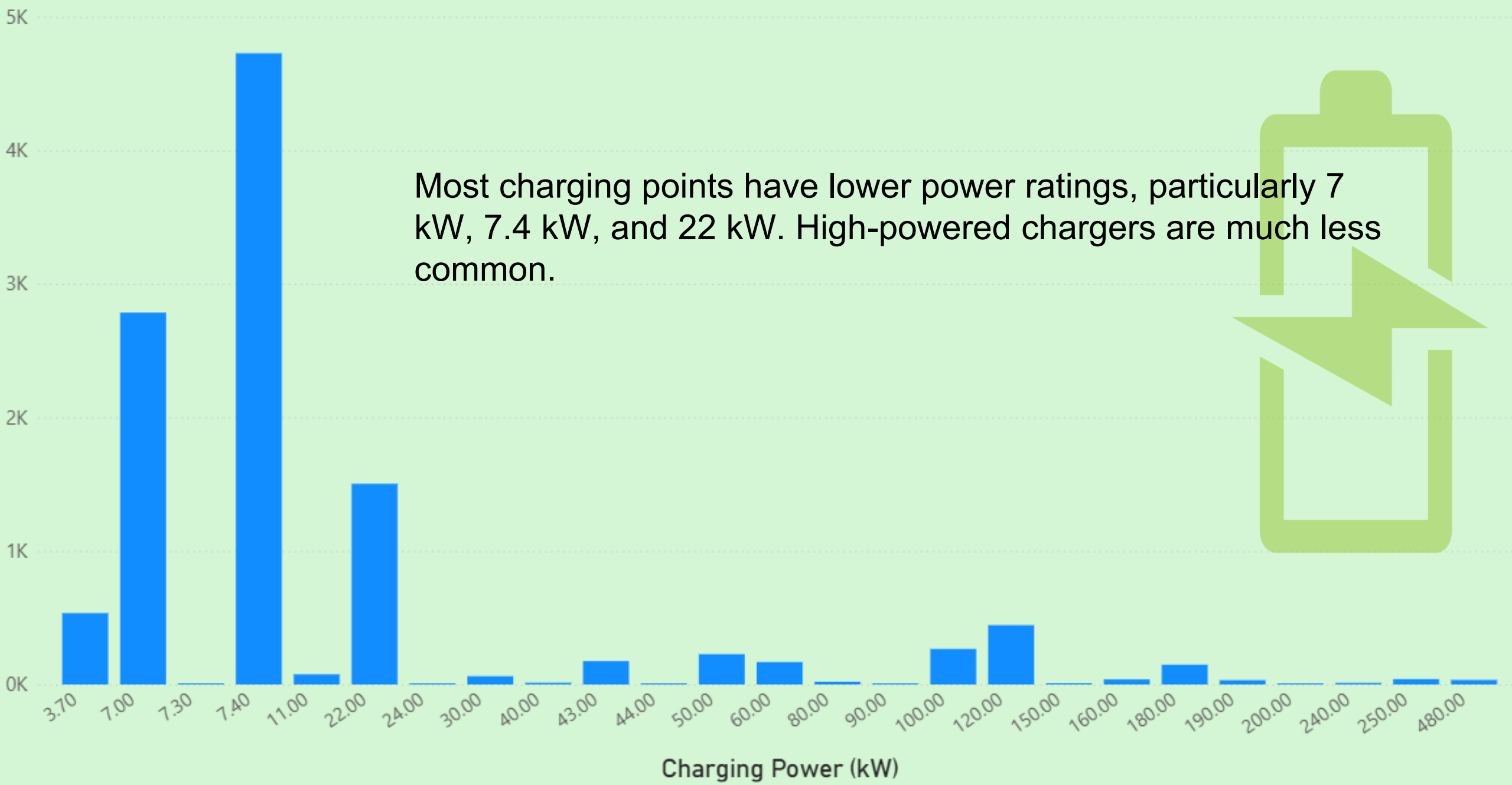


Distribution of EV Charging Points by Region



The Central and West regions together account for approximately 55% of all charging points, while the North has the smallest share at 12.18%.

Distribution of EV Charging Points by Power Rating



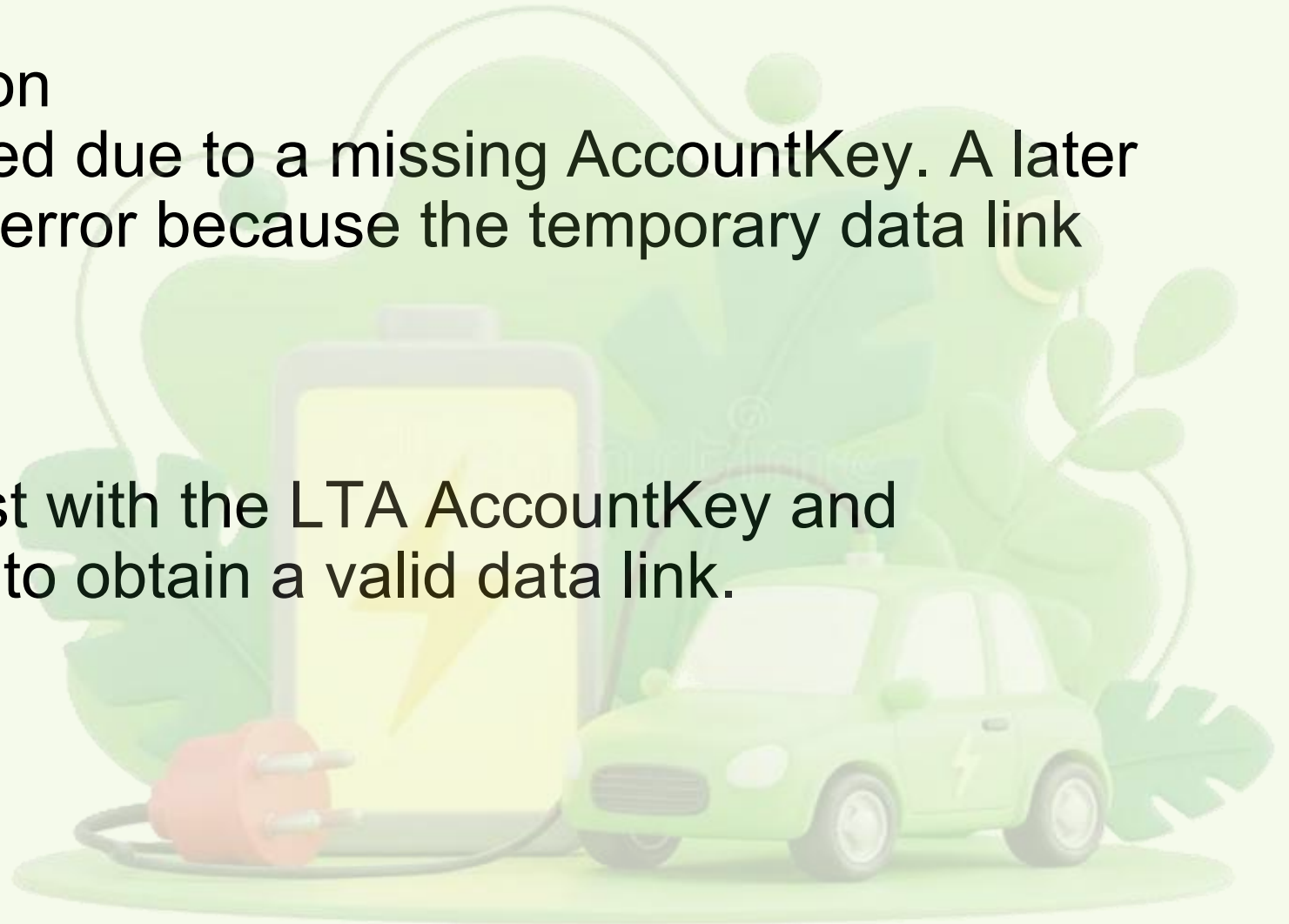
Challenge 1

Problem: API connection

The request initially failed due to a missing AccountKey. A later attempt returned a 403 error because the temporary data link had expired.

Solution:

Sent a new GET request with the LTA AccountKey and application/json header to obtain a valid data link.



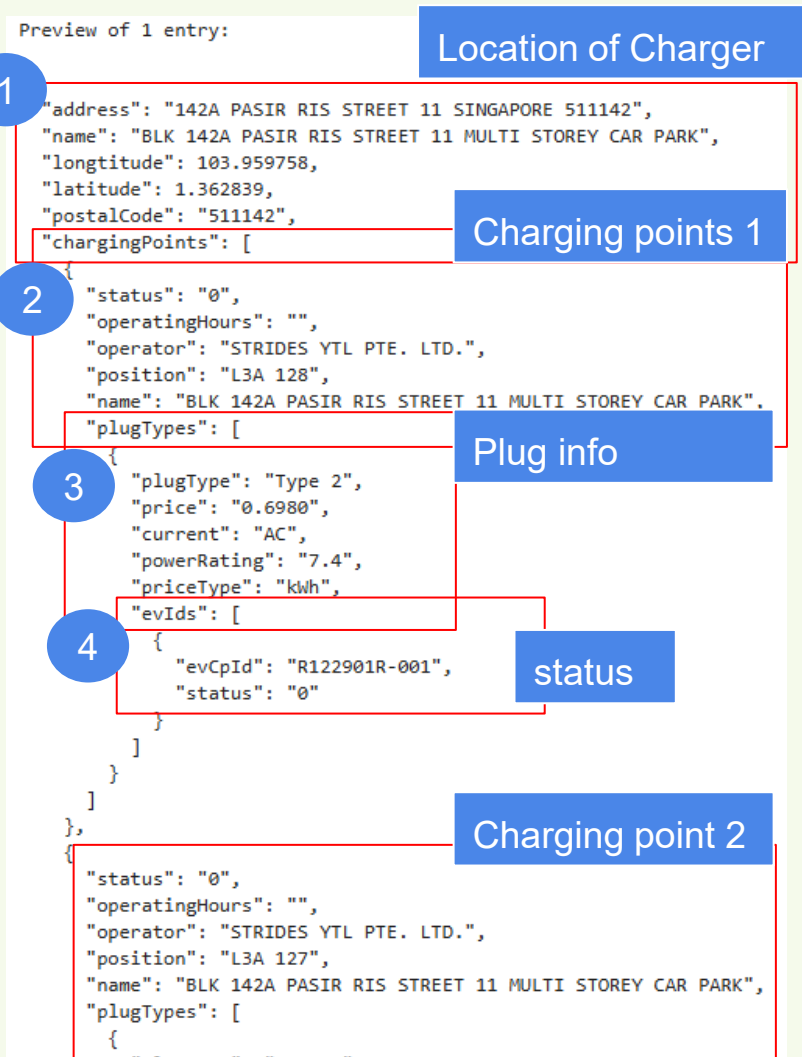
Challenge 2

Problem: Flattening four nested levels

Solution:

- Approach 1: Split into 4 separate tables
- Approach 2: Loops to extract and combine
- Approach 3: `pd.json_normalize()` X3 + `.to_dict()`

Selected: Approach 3

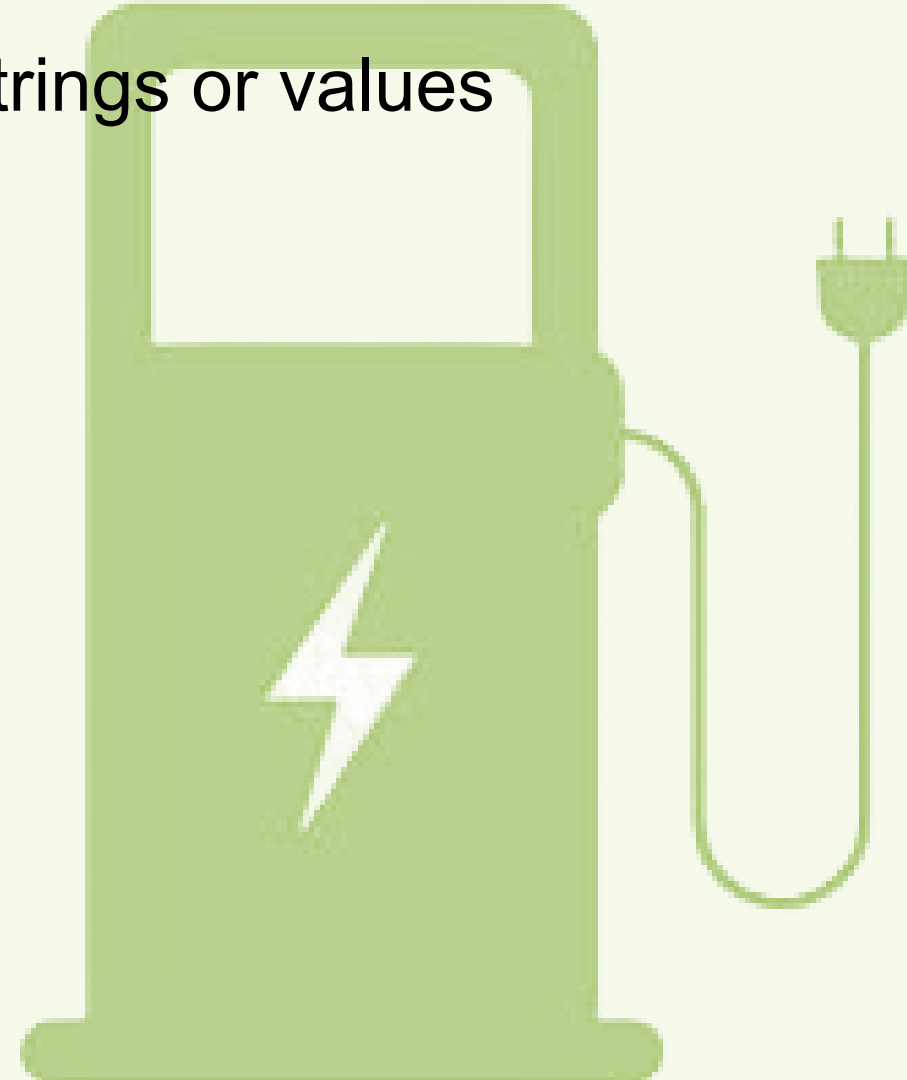


Challenge 3

Problem: NULL values
isnull() and isna() could not detect empty strings or values
containing only spaces.

Solution:

- value_counts()
- Replaced " with np.nan



Challenge 4

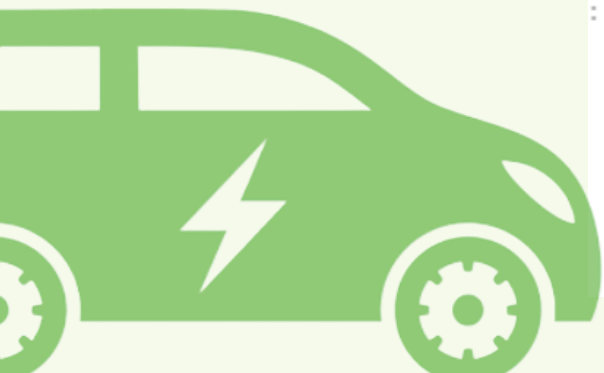
Problem: Data-Type Incompatibility

The data could not be loaded into the PostgreSQL table because the DataFrame and database column types were incompatible.

Solution:

Converted the DataFrame columns to the appropriate data types before loading the data into PostgreSQL. Creating the table beforehand provides greater control over its schema, constraints and data types.

```
: # converting to a more appropriate data type for each column  
  
df_main.postal_code = df_main.postal_code.astype(int)  
df_main.chargingpoint_status = df_main.chargingpoint_status.astype(int)  
df_main.price = pd.to_numeric(df_main.price, errors='coerce')  
df_main.evid_status = pd.to_numeric(df_main.evid_status, errors='coerce')
```



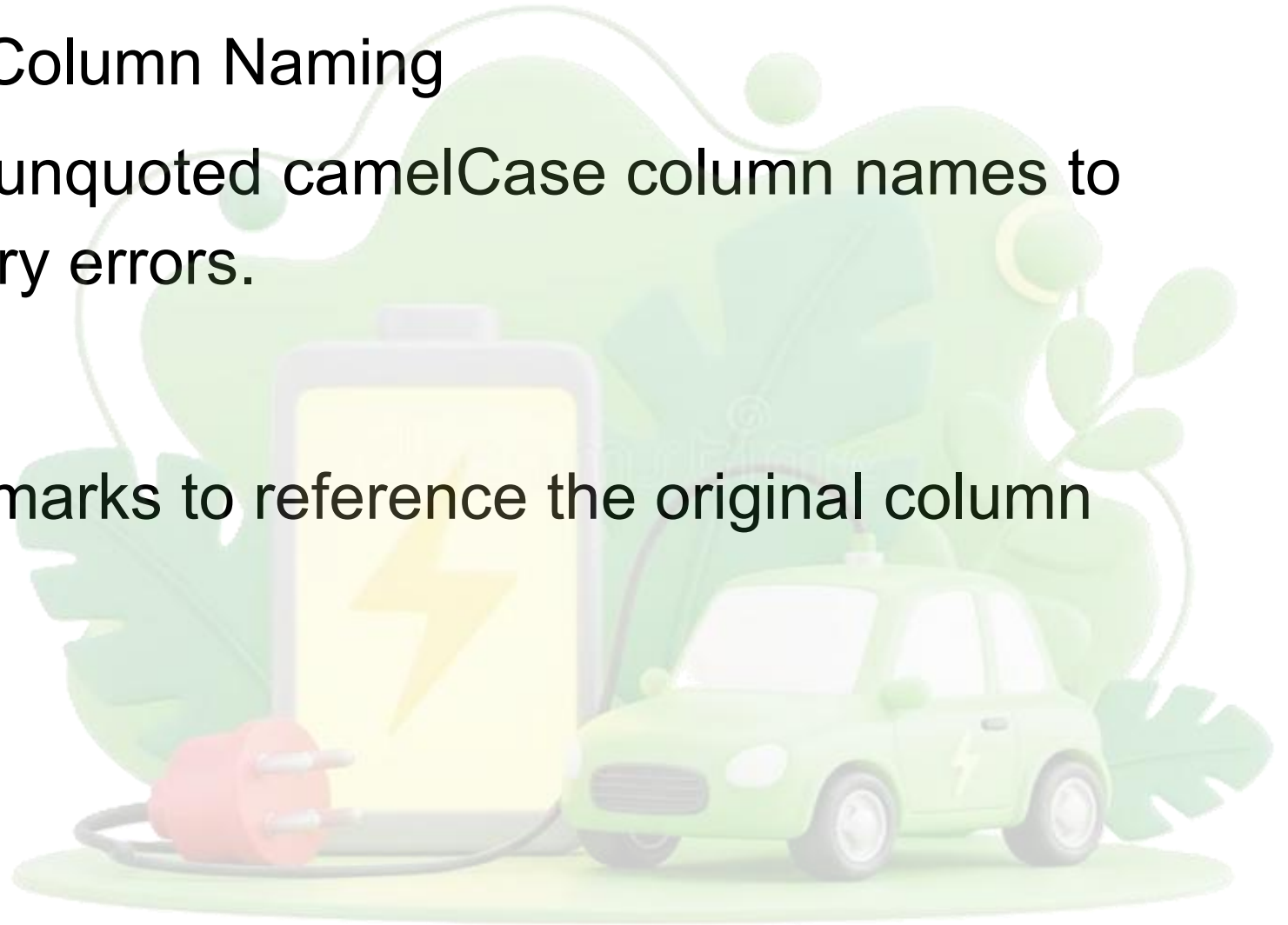
Challenge 5

Problem: Inconsistent Column Naming

PostgreSQL converted unquoted camelCase column names to lowercase, causing query errors.

Solution:

Used double quotation marks to reference the original column names.

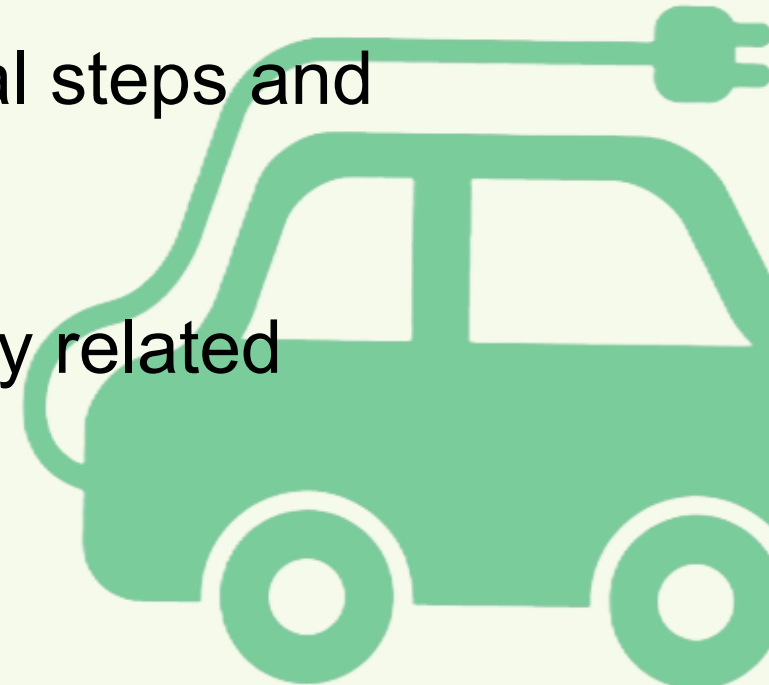


Limitations

- The data is a snapshot and does not support historical trend analysis.
- Missing price and price-type values may affect some analyses.
- Recent charger installations or maintenance may not yet be reflected.
- Regional classifications rely on a community-created GitHub mapping rather than an official government dataset.

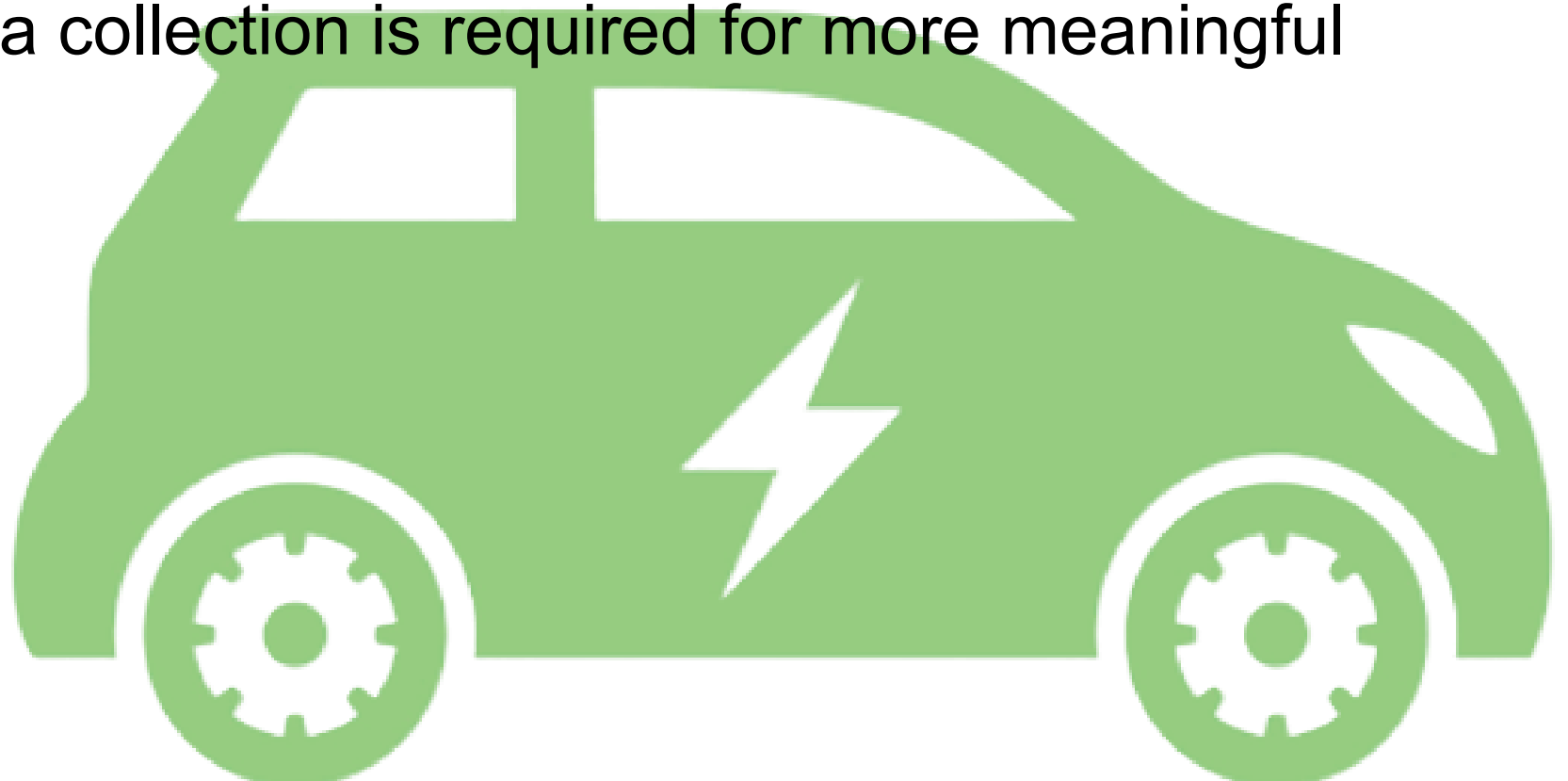
Areas for Improvement

- Automate validation checks after each transformation stage.
- Confirm data types, column names and database design before loading the data.
- Streamline the ETL process to reduce manual steps and improve consistency.
- Agree on the project scope earlier and identify related datasets before development begins.



Conclusion

The project successfully transformed complex EV charging-point data into a structured PostgreSQL database. The resulting dataset supports analysis of charging operators, power ratings and regions. However, regular data collection is required for more meaningful historical analysis.



Thank You
